

DMCA — Joint Collaboration Outline

Distributed AAS-based Digital Twin for Constraint-Driven Edge AI Orchestration

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Working document — for joint review

CONTEXT

DMCA (Dynamic Model-Context Alignment) is a five-stage closed-loop framework for adaptive selection and re-alignment of ML models on edge AI deployments, driven by Digital-Twin-encoded asset constraints. Phase 1 has been implemented and empirically validated as part of the author's M.Sc. thesis (Politecnico di Torino × Beihang, defence scheduled July 2026), using a JSON-serialised AAS Type 1 as the DT substrate.

Following an exchange in May 2026, Dr. A. Belfadel (IRT SystemX) outlined a natural extension of Stage 1, quoted below, which the present note proposes to develop jointly as a post-thesis collaboration.

“As AAS follows a distributed architecture, different parts of the AAS-based DT can be deployed across cloud, fog, and edge nodes [...]. We could implement a constraint-driven ML model selection through a multi-criteria selection approach based on specific submodels holding the actual status of each node or device.”

PROPOSED JOINT SCOPE (STAGE 1 EXTENSION)

Move Stage 1 of DMCA from a static JSON AAS representation to a **distributed AAS architecture** in which:

- submodels are dynamically populated with runtime status (latency, load, memory headroom, drift indicators) of each cloud/fog/edge node;
- the multi-criteria selection step (currently TOPSIS over a static registry) is exposed as a **DT service** that consumes these submodels;
- the framework is positioned conformantly to **IEC 63278 (CDV)** and **ISO 23247 / ISO/IEC 30173**.

This extension does not replace Phase 1 of the thesis; it generalises it from single-asset / static AAS to multi-node / dynamic AAS.

TENTATIVE FOUR-PARTY ROLE MAP

Contributor	Institution	Proposed role
K. Omede	Polito × Beihang	Lead author and executor; DMCA core (TOPSIS, drift detection v3.1.1, benchmark); integration of the distributed-AAS extension.
Prof. A. Simeone	Politecnico di Torino	Senior supervision (Italian side); methodological framing; access to industrial validation context.
Dr. A. Belfadel	IRT SystemX	Technical collaborator; distributed AAS architecture; submodel design as DT service; possible IRT industrial use case (to be discussed).
Prof. N. Anwer	Paris-Saclay / LURPA	Senior supervision (French side); standards positioning (ISO/IEC SC 41, IEC 63278); strategic guidance and venue access.

TENTATIVE DELIVERABLES

- **Primary:** a co-authored journal paper, target venue *IEEE Transactions on Industrial Informatics* (or equivalent), submission window Q4 2026 / Q1 2027.
- **Secondary:** a possible workshop or short paper presented within a CIRP-affiliated venue, scoped jointly with Prof. Anwer.
- **Tertiary (optional):** a continued Phase 2 research line should the initial collaboration prove productive — to be discussed only after Phase 1 extension is consolidated.

OPEN POINTS FOR JOINT DEFINITION

- Scope and depth of the **distributed AAS implementation** (e.g. Eclipse BaSyx integration, level of submodel dynamism).
- Availability of an **industrial use case from IRT SystemX** for validation, and corresponding data-sharing or NDA arrangements.
- Final **authorship order** and senior-author convention across the two institutions.
- Tentative **target submission date** and review schedule, anchored on post-thesis availability (from August 2026).

PROPOSED NEXT STEP

A 30–45 minute joint videoconference in the next two to three weeks, to align on the open points above and confirm the collaboration scope before any implementation work is committed.

Working draft — not for redistribution outside the proposed collaboration. Contact: kevinomede2002@gmail.com